CSS-Mamba: Competitive Sparse-Selection Mamba Network for Hyperspectral–LiDAR Joint Classification

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***Abstract*—Hyperspectral imaging (HSI) and light detection and ranging (LiDAR) data provide highly complementary information in terms of fine-grained spectral characterization and 3-D structural description, which is crucial for accurate land-cover classification in complex scenes. However, most existing multimodal fusion methods adopt isomorphic feature extraction and weighted aggregation strategies, which fail to explicitly model the intrinsic differences between spectral sparsity in HSI and geometric sparsity in LiDAR. Moreover, multiscale fusion is often performed without explicit competitive selection mechanisms, leading to redundant feature accumulation and insufficient responses at discriminative scales. To address these limitations, this article proposes a competitive sparse-selection Mamba network (CSS-Mamba) for hyperspectral–LiDAR joint classification. The proposed framework integrates modality-specific sparse modeling with competitive multiscale feature selection in a unified architecture. Specifically, a multiscale spectral–spatial sparse encoding (MS-SSSE) module and a multiscale geometric structure sparse encoding (MS-GSSE) module are designed to separately model the spectral sparsity of HSI and the structural sparsity of LiDAR. Learnable soft-thresholding operators are embedded into the network to impose sparsity constraints at the architectural level, enabling modality-differentiated feature representations. Furthermore, a competitive sparse selection (CSS) module is introduced, which combines visual selective state-space blocks (VSS blocks) with spatial competitive selection (SCS) to establish explicit competition among multiscale features, adaptively enhancing discriminative scales while suppressing redundant ones. Extensive experiments conducted on three public benchmark datasets, including Trento, Houston 2013, and MUUFL, demonstrate that the proposed CSS-Mamba consistently outperforms representative convolutional neural network (CNN)-, Transformer-, and Mamba-based multimodal fusion methods in terms of overall accuracy, average accuracy, and Kappa coefficient. The results verify the effectiveness and robustness of the proposed modality-aware sparse modeling and competitive scale selection strategy, and highlight its potential as a general and efficient framework for multimodal remote sensing data fusion.**

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I. Introduction

In THE context of rapid advancements in multisource remote sensing observation technologies, the integration of hyperspectral imaging (HSI) and light detection and ranging (LiDAR) data has emerged as a pivotal research direction in the domain of land-cover classification [1] and feature identification. HSI provides continuous spectral response information [2], [3] [4], thereby enabling the characterization of the reflectance properties of objects across a range of wavelengths [5], [6]. This, in turn, facilitates effective differentiation of materials and biochemical compositions. LiDAR technology has been demonstrated to capture the 3-D geometric structure and elevation distribution of the Earth's surface with a high degree of precision [7], thus compensating for the spatial limitations that are inherent to hyperspectral data. The complementary nature of these two technologies enables HSI–LiDAR fusion to enhance the accuracy of complex feature recognition and classification across spectral, spatial, and structural dimensions [8]. In recent years, significant progress has been witnessed in multisource fusion models, driven by rapid advancements in deep learning technologies. From early convolutional neural networks (CNNs) [9], [10] to attention-based Transformers [11], [12] [13], and the recently emerging state-space models (SSMs) and their efficient implementation Mamba [14], [15], these models have achieved breakthroughs in feature extraction capabilities, contextual modeling scope, and fusion depth. However, as model structures become more complex, the potential for enhancement of performance becomes more limited. The enhanced representational capabilities of these models have not been shown to result in significant gains in classification accuracy. This finding suggests that contemporary multisource fusion models continue to encounter significant limitations in their feature representation mechanisms and fusion strategies.

In order to surmount the aforementioned impediments, this study revisits the foundational principles of sparse representation theory in order to reexamine the mathematical essence of HSI-LiDAR joint classification. The primary information of hyperspectral imagery is derived from the spectral domain. The high-dimensional spectral curves of the latter contain rich chemical and biological signatures, exhibiting

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pronounced spectral sparsity [16], [17]. This is characterized by highly correlated yet subtly distinct reflectance responses across different bands. In contrast, LiDAR data's discriminative information is concentrated in spatial structure and elevation variations [18], [19], with sparsity primarily manifesting in geometric domains such as building edges, canopy morphology, and terrain undulations. The physical significance, signal-to-noise characteristics, and spatial distribution features of these two modalities exhibit fundamental differences. The majority of contemporary HSI-LiDAR fusion networks utilize identical convolutional encoding structures or shared parameter spaces [20], [21], thereby mapping disparate modal features into a unified feature domain. While this "isomorphic processing" formally achieves modal unification, it semantically weakens modal independence and complementarity. This approach frequently results in the loss of spectral detail and the excessive smoothing of geometric information, which, in turn, leads to redundant and ambiguous feature representations. Furthermore, in complex urban environments or vegetated areas, spectral and structural signals exhibit markedly different statistical distributions. In the absence of modality-specific constraints and sparse representation mechanisms, models frequently exhibit feature bias, training instability, and inadequate generalization capabilities. Consequently, the primary challenge in multisource fusion is the construction of a sparse feature modeling mechanism that preserves modality specificity while achieving complementary enhancement.

Moreover, extant methodologies continue to manifest substantial deficiencies with regard to the selection and competitive mechanisms of multiscale features. Land-cover objects vary considerably across spatial scales, with feature patterns spanning multiple levels. These levels range from low-frequency structures, such as large building complexes, areas of bare land, and terrain undulations, to high-frequency details, including fine-grained vegetation, road edges, and small structures. Multiscale fusion methods of a traditional nature generally implement parallel convolutions [22], pyramid feature aggregation [23], [24], or multihead attention mechanisms [25], [26], [27] in order to perform weighted fusion across multiple-scale features. For instance, in dense urban scenes, parallel convolution-based methods tend to conflate road edges with adjacent building facades across scales, while pyramid aggregation approaches in agricultural landscapes often blur fine-grained boundaries between spectrally similar crops such as vineyards and orchards due to indiscriminate scale weighting. While these approaches have the capacity to integrate multiscale context to a certain degree, the majority are deficient in the area of explicit competitive modeling mechanisms. The weight distribution across scale features frequently exhibits a tendency to smooth, thereby impeding models from accentuating dominant scale features in complex scenes. A number of studies have sought to implement gated or attention-weighted structures with a view to enhancing discriminative power by learning saliency weights across scales. However, in the absence of sparse constraints, these approaches are susceptible to "all-in" strategies, leading to feature redundancy and ambiguous discrimination. Furthermore, the simple stacking of multiscale information increases computational complexity and redundant noise, causing confusion in fusion results at boundaries or transitional features. This deficiency persists in existing state-space model-based fusion methods, which adopt implicit aggregation strategies during multiscale fusion without establishing explicit competitive relationships among scale features, causing redundant scale responses to accumulate rather than being adaptively suppressed—a problem that is particularly pronounced at boundaries between spectrally similar or spatially confusable categories. Consequently, there is an urgent necessity for a multiscale fusion mechanism with competitive selection capabilities. This mechanism should enable adaptive sparse selection among features at different scales, allowing the model to focus on the most discriminative scale information in complex scenes.

In order to address these challenges, this article proposes a competitive sparse-selection Mamba network (CSS-Mamba). The fundamental concept of this framework entails the integration of modality-specific sparse modeling with a scale-competition fusion mechanism. During the process of feature extraction, distinct modules are designed for each modality's imaging mechanism. These modules include the multiscale spectral–spatial sparse encoding (MS-SSSE) module and the multiscale geometric structure sparse encoding (MS-GSSE) module. MS-SSSE employs spectrum-guided deep separable convolutions and channel attention mechanisms to suppress redundancy and enhance local spatial patterns in the spectral domain (hyperspectral data). MS-GSSE focuses on geometric contours and morphological features in LiDAR data through directional strip convolutions. Furthermore, a learnable soft-thresholding layer has been embedded within both MS-SSSE and MS-GSSE modules for the purpose of sparsity enhancement. This layer simulates a single step of the iterative shrinkage-thresholding algorithm (ISTA), achieving elementwise sparsity through learnable threshold parameters. This process implements modality-specific sparsity constraints at the architectural level. During the fusion stage, this article introduces a competitive sparse selection (CSS) module. The integration of a visual selective state-space block (VSS block) facilitates the capture of long-range dependencies while establishing explicit competitive relationships across multiple scales through the hierarchical application of spatial competitive selection (SCS) blocks. In contrast to conventional fixed-weight or averaging fusion methodologies, this competitive selection constraint employs a dynamic suppression of redundant scales, thereby enhancing discriminative scale responses based on feature characteristics through a sparse weight allocation mechanism. This enables the model to adaptively focus on optimal scale combinations, thereby improving classification accuracy and robustness.

The main contributions of this article can be summarized as follows.

1) A competitive sparse-selection Mamba framework is proposed for hyperspectral–LiDAR joint classification. Unlike existing multimodal fusion methods that rely on homogeneous feature extraction and implicit aggregation, the proposed CSS-Mamba introduces an explicit competitive sparse-selection

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mechanism within a state-space modeling framework. This design enables effective modeling of long-range contextual dependencies while selectively emphasizing discriminative multiscale features, providing a new perspective for multimodal remote sensing data fusion.

2) Modality-specific sparse encoding strategies are developed to explicitly characterize the intrinsic differences between HSI and LiDAR data. By designing the multiscale spectral–spatial sparse encoding (MS-SSSE) module for HSI and the MS-GSSE module for LiDAR, the proposed method embeds learnable soft-thresholding operators into the network architecture. This allows spectral sparsity and geometric sparsity to be modeled in a modality-aware manner, leading to more compact and discriminative feature representations.

3) A CSS module is introduced to enhance multiscale feature discrimination. The proposed CSS module establishes explicit competition among multiscale features by integrating VSS blocks with SCS. This mechanism adaptively strengthens informative scales while suppressing redundant responses, alleviating the common issue of feature redundancy in multiscale fusion and improving robustness in complex scenes.

4) Extensive experiments on multiple benchmark datasets demonstrate the effectiveness and generalization ability of the proposed approach. Experimental results on the Trento, Houston 2013, and MUUFL datasets show that CSS-Mamba consistently outperforms representative CNN-, Transformer-, and Mamba-based multimodal fusion methods in terms of overall accuracy, average accuracy, and Kappa coefficient, validating its robustness and suitability for hyperspectral–LiDAR joint classification tasks.

The remainder of this article is organized as follows. Section II provides a review of the related work. Section III introduces the proposed framework and its key components. Section IV presents a comprehensive overview of the experimental dataset, implementation details, and performance analysis. Finally, Section V puts forward several potential avenues for future research.

II. Related Work

A. CNN-Based Approach for Multisource Remote Sensing Data Fusion

Early studies on joint classification using HSI and LiDAR primarily relied on CNNs, whose core concept involves parallel extraction and fusion of multimodal features through multibranch architectures.

Such methods typically employ feature-level concatenation, weighting, or attention-guided interactions at intermediate layers to enhance information exchange and complementarity between modalities. For instance, Khordadzadeh et al. [28] proposed a multisource fusion approach based on extended morphological profiling (EMP). This method combines spectral information from HSI with elevation features from LiDAR, feeding the integrated data into a support vector machine (SVM) classifier to achieve fusion at the decision level. This approach effectively leverages LiDAR's structural and topographic elevation information, thereby mitigating the issue of insufficient spectral discrimination among similar categories in hyperspectral imagery. In the field of remote sensing, the attention-based multimodal fusion network (FusAtNet) framework designed by Mohla et al. [29] has been found to employ a dual-branch CNN architecture, a sophisticated computational model, to process HSI and LiDAR data separately. The integration of cross-attention modules at intermediate layers facilitates explicit feature interaction between modalities, thereby enhancing the semantic consistency of fused features. The dynamic cross-modal feature interaction network (DCM-Net), as conceptualized by Lin et al. [30], is an innovative framework that aims to enhance spatial and channel feature representations through the utilization of bilinear attention blocks. It has been demonstrated that a dynamic routing mechanism is capable of adapting to the terrain in order to determine fusion paths. This has been shown to have a significant effect on classification performance in complex terrain areas.

In addition to the aforementioned approaches, an increasing number of studies have sought to incorporate multiscale convolutional kernels and attention mechanisms into CNN architectures. This is with a view to accommodating both the spectral continuity of HSI and the spatial structural features of LiDAR. For instance, Ge et al. [31] employed a spatioelevation–spectral convolutional feature extraction module, utilizing an enhanced multiscale convolutional block to extract features from both HSI and LiDAR, thereby ensuring discriminative capability and comprehensiveness under diverse land-cover conditions. Zhou et al. [32] devised a multiscale segmentation-recombination collaborative fusion approach, extracting discriminative features through parallel multiscale hierarchical inverted pyramid modules while employing self-attention mechanisms to model global relationships across modalities. Pan et al. [22] proposed a position-channel collaborative attention network, extracting spatial, spectral, and elevation features via a multiscale architecture and utilizing collaborative attention modules to adaptively enhance feature representation, thereby mitigating semantic discrepancies between heterogeneous data. However, CNN-based methods adopt isomorphic convolutional paths to process HSI and LiDAR features, failing to differentiate between the spectral sparsity of HSI and the geometric sparsity of LiDAR. Applying identical convolutional architectures to two modalities with fundamentally distinct physical properties frequently leads to information redundancy and feature aliasing, thereby preventing the explicit modeling of sparse structures. Overall, while CNNs demonstrate high efficiency in multisource feature extraction and fusion, their local modeling nature and fixed receptive field limit their capture of long-range dependencies and modal sparsity. This limitation has paved the way for subsequent research opportunities, notably the introduction of Transformers and SSMs.

B. Transformer-Based Approach for Multisource Remote Sensing Data Fusion

The introduction of Transformers has provided powerful global modeling capabilities for multisource remote sensing data fusion. The self-attention mechanism employed by the model effectively captures long-range dependencies and global

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contextual relationships, thereby overcoming the limitations of CNNs in terms of local receptive fields and feature interactions.

Early Transformer variants primarily focused on single-modal HSI classification, such as SpectralFormer [33] and the spectral–spatial feature tokenization Transformer (SSFTT) [34], achieving long-range information modeling through spectral attention and spatiotemporal attention. As research progressed, the Transformer architecture underwent rapid expansion into multimodal scenarios, becoming a pivotal direction for multisource data fusion. Roy et al. [13] proposed the multimodal fusion Transformer (MFT), which fuses patch tokens from HSI and LiDAR through a multihead cross-patch attention (mCrossPA) mechanism, thereby achieving hierarchical semantic representations. Yao et al. [35] developed the extended visual Transformer (ExViT), which employs position-shared ViT branches to independently encode multimodal patches. A cross-modal attention (CMA) module facilitates pixel-level spatial correlation interactions, enhancing the model's ability to consistently represent heterogeneous data. Roy et al. [36] further proposed the Cross-HL Transformer, in which LiDAR patches are employed as queries (Q) and HSI patches as keys (K) and values (V), thereby enabling modal mutual guidance within the self-attention architecture. The hierarchical CNN–Transformer (HCT) model proposed by Zhao et al. [37] integrates the local modeling strengths of CNN with the global representation capabilities of Transformer, thereby achieving multilayer feature fusion through cross-token attention. Wang et al. [38] developed a dual-branch cross-Transformer network for land-cover classification tasks, emphasizing modality-specific feature extraction and cross-branch semantic alignment.

In recent years, the Transformer architecture has seen continuous expansion in its applications for multimodal remote sensing fusion. The cross-modal feature aggregation and enhancement network (CMAENet) proposed by Zhang et al. [39] employs multilayer cross-modal aggregation blocks with the objective of strengthening high-level semantic features. This, in turn, leads to an improvement in the discrimination capability of similar categories in urban scenes. Ni et al. [40] pioneered the multiscale head selection Transformer (MHST), a pioneering approach that delves into the intricate realm of nonredundant feature extraction across the intricacies of spatial and spectral dimensions. Sun et al. [41] proposed the multiscale 3D–2D Hybrid CNN feature extraction and multisource data lightweight fusion network (M2FNet), introducing a feature enhancement module based on mathematical morphology and a dilated convolutional Transformer encoder (DConvformer) to achieve multiscale deep feature learning. Furthermore, Wang et al. [42] proposed the convolutional interaction Transformer (CITNet), which jointly utilizes local–global Transformer (LGTM) and optimized convolutional cross-attention (OCA) to extract deep local–global features, enhancing spatial modeling capabilities between semantic tokens.

Transformer-based approaches have been demonstrated to achieve long-range dependency modeling through global self-attention, thus exhibiting superior performance in the domain of multimodal data fusion tasks. Nevertheless, their elevated computational complexity imposes constraints on their applicability in large-scale hyperspectral scenarios. Furthermore, Transformer-based methods tend to yield oversmoothed attention weight distributions and lack explicit competitive selection mechanisms. Consequently, they are prone to retaining redundant features during multiscale fusion while overlooking the intrinsic differences in the sparse distributions between HSI and LiDAR. The critical challenge for advancing Transformer-based fusion models lies in the introduction of interpretable sparsity and competitive mechanisms while preserving global modeling capabilities.

C. SSM-Based Approach for Multisource Remote Sensing Data Fusion

In recent years, SSMs [43] have emerged as a novel sequential modeling paradigm. Due to their linear time complexity and their capacity for modeling long sequences, they have gradually become a new direction in the field of remote sensing image analysis. The representative model Mamba achieves efficient capture of long-range dependencies by introducing state-space dynamic equations and selective gating mechanisms while significantly reducing the computational overhead of Transformers. In contradistinction to conventional attention mechanisms, SSM enables feature-to-feature information transfer through implicit state propagation, thus conferring distinct advantages in terms of modeling efficiency and memory consumption. This characteristic demonstrates the considerable potential of the Mamba architecture for processing large-scale hyperspectral and multisource data.

Pan et al. [22] proposed the multimodal fusion Mamba network (M2FMNet), employing spatial–spectral adaptive Mamba modules and elevation-enhanced Mamba modules. The scanning paths were specifically designed for HSI and LiDAR data to achieve pixel-level fine-grained fusion. Li et al. [44] introduced the semisupervised Mamba framework (Semi-Mamba), which integrates cross-modal fusion modules with semisupervised learning mechanisms. This approach enhances the performance of model generalization by incorporating unlabeled samples. Gao et al. [15] developed the multiscale fusion Mamba (MSFMamba) architecture, which combines spatial Mamba (SpaMamba), spectral Mamba (SpeMamba), and fusion Mamba (FusMamba) modules to model multiscale spatial–spectral–elevation feature variations. This achieves multilevel alignment of heterogeneous data. Liao et al. [14] proposed the HL-Mamba network, which integrates LiDAR gradient extraction with spectral-domain enhancement strategies. This work demonstrates the crucial role of LiDAR gradient features in joint classification. Li et al. [45] further introduced the multiscale hybrid convolutional Mamba (MHCMamba), which fuses convolutional operations with state-space dynamic modeling to balance local structure learning and global dependency capture. This approach has been shown to achieve outstanding results in multimodal classification.

In conclusion, SSM-based approaches provide a novel research direction for multisource remote sensing data fusion,

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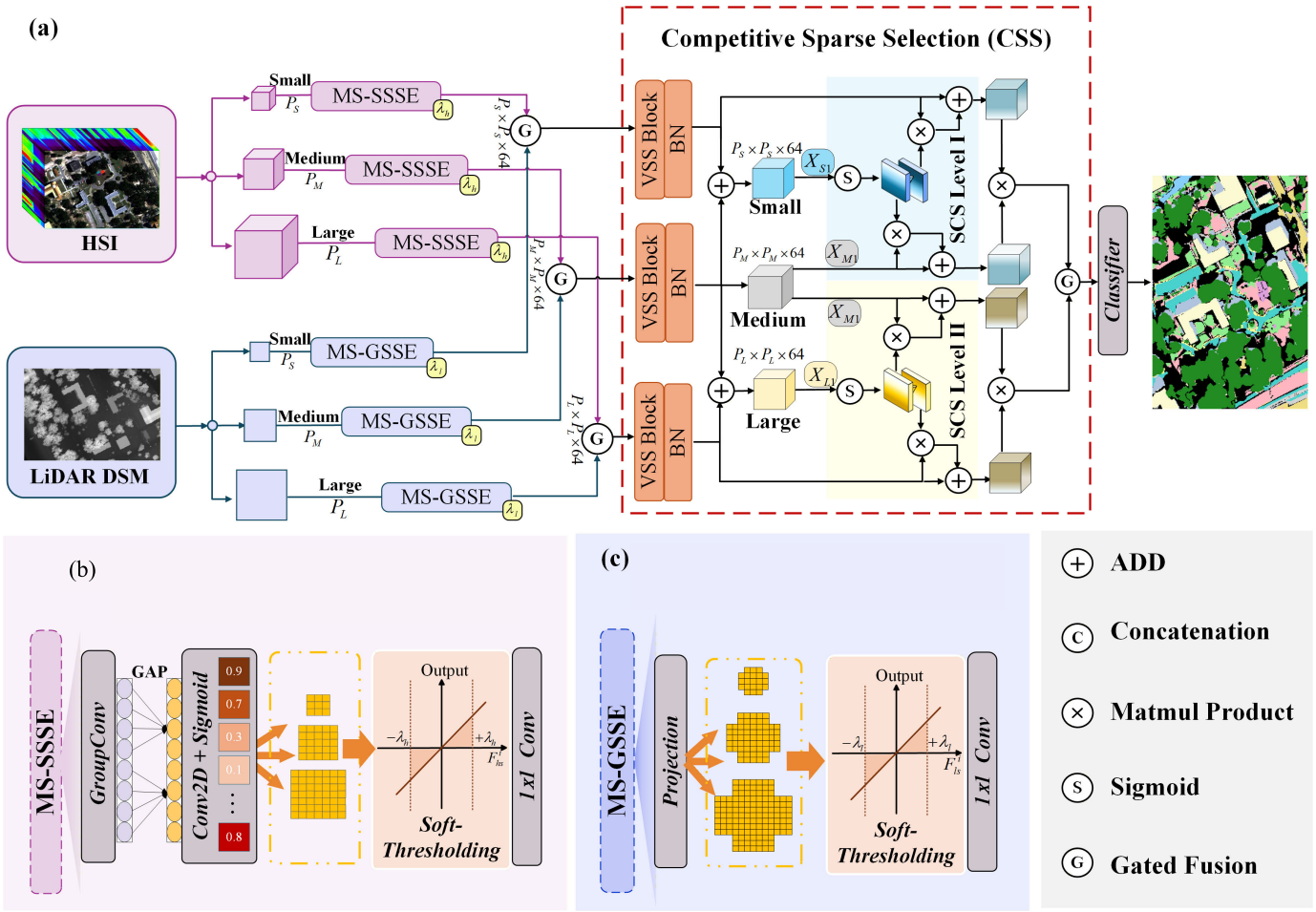


Fig. 1. (a) Framework of the proposed CSS-Mamba. (b) MS-SSSE. (c) MS-GSSE.

enabling long-range dependency modeling and global context capture at relatively low computational cost. However, existing Mamba-based methods still employ a unified state-space structure during the modality modeling stage, without introducing differentiated designs tailored to the specific sparse characteristics of HSI and LiDAR. Furthermore, they similarly lack an explicit competitive selection mechanism during the multiscale fusion process. In order to address these shortcomings, the proposed CSS-Mamba framework employs guided sparse learning and a competitive scale selection module. This approach preserves the computational advantages of the Mamba architecture while enabling modality-specific sparse modeling and multiscale dynamic fusion. Consequently, it significantly enhances the accuracy and robustness of HSI–LiDAR joint classification.

III. Methodology

An overview of the proposed CSS-Mamba framework is presented in Fig. 1(a). CSS-Mamba consists of MS-SSSE for HSI, MS-GSSE for LiDAR, and a CSS module, which are jointly optimized in an end-to-end manner.

Specifically, given hyperspectral patches and their corresponding LiDAR patches at three spatial scales (small, medium, and large), the MS-SSSE and MS-GSSE modules extract modality-specific sparse features in parallel for each scale. The resulting HSI and LiDAR feature pairs at each scale are then fused via a gated fusion mechanism, yielding three scalewise fused representations. These representations are subsequently fed into the CSS module, which performs CSS across scales to capture long-range dependencies and adaptively enhance discriminative scale responses. Finally, the aggregated output is passed to a classifier to produce the category prediction.

A. MS-SSSE Module

The MS-SSSE module employs a four-step process to perform modality-specific spectral sparse modeling. As demonstrated in Fig. 1(b), the module attains mode-specific spectral sparse modeling. The MS-SSSE module is predicated on the physical prior that HSI data features rich spectral dimensions and spectral-domain sparsity. First, modality-specific structural features are extracted in order to identify discriminative characteristics. Subsequently, an embedded sparse coding layer is employed to enforce sparsity, thereby achieving "native feature sparsity." The present module employs a four-step process to accomplish this objective: the module is divided into four stages: spectral subspace decomposition, discriminative

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channel selection, spatial pattern modeling and sparse solving, and feature reconstruction.

1) *Spectral Subspace Decomposition:*

Given a hyperspectral image, $X_H^i \in \mathbb{R}^{P_i \times P_i \times B}$, where $i \in \{s, m, l\}$ represents three different scales. Initially, the spectral subspace decomposition must be executed by means of 1-D grouped convolutions with a grouping size of $G = B/4$

$$F_{sp}^i = \text{GroupConv1D}\left(X_H^i, G\right) \tag{1}$$

where $F_{sp}^i \in \mathbb{R}^{P_i \times P_i \times B}$ denotes the results of the spectral subspace decomposition. In contrast to conventional fully connected 1-D convolutions, the grouped convolution architecture decomposes the B-dimensional spectral features into G independent subspaces, with each group capturing a distinct set of spectral response patterns. This configuration enforces structural sparsity, whereby each convolution kernel group constitutes a "subdictionary" responsible for learning only a subset of these patterns.

2) *Adaptive Selection of Discrimination Spectral Channels:*

We introduce a lightweight spectral attention mechanism that adaptively identifies the spectral features most crucial for classification tasks. This mechanism enables spectral features to guide spatial features

$$W_{sp}^i = \sigma\left(\text{Conv}_{1\times1}^{\text{up}}\left(\text{ReLU}\left(\text{Conv}_{1\times1}^{\text{down}}\left(\text{GAP}\left(F_{sp}^i\right)\right)\right)\right)\right) \tag{2}$$

$$H_{attn}^i = W_{sp}^i \odot H_{sp}^i \tag{3}$$

where GAP(•) denotes global average pooling (GAP), $\text{Conv}_{1\times1}^{\text{down}}$ and $\text{Conv}_{1\times1}^{\text{up}}$ represent dimensionality reduction and increase using 1×1 convolutions, respectively, σ is the Sigmoid activation function, W_{sp}^i is the learned channel attention weights, and H_{attn}^i the attention-weighted spectral features.

3) *Spatial Pattern Modeling and Sparse Solving:*

Following the extraction of discriminative spectral features F_{attn}^i, the deeply separable convolution (DSConv) algorithm is employed [46] with kernel sizes of $k \times k$ (where $k \in \{3, 5, 7\}$ corresponds to small, medium, and large scales, respectively). This process serves to enhance local spatial patterns. Initially, deep convolution is performed

$$F_{dw}^i = \text{DWConv}_{k\times k}\left(F_{attn}^i\right) \tag{4}$$

where F_{dw}^i is the features that are capable of capturing spatial patterns. Subsequently, to explicitly implement sparse coding at the architectural level, a key sparse decoding layer is introduced. We apply the soft-thresholding operator $S_{\lambda_h}(\cdot)$ to the feature map F_{dw}^i

$$F_{sph}^i = S_{\lambda_h}\left(F_{dw}^i\right) \triangleq \text{sign}\left(F_{dw}^i\right) \cdot \max\left(|F_{dw}^i| - \lambda_h, 0\right) \tag{5}$$

where F_{sph}^i are the sparsified features and λ_h is a learnable threshold parameter.

4) *Feature Reconstruction:*

The sparse features obtained are then fed into a pointwise convolution, which acts as a "decoder" in this context, responsible for mapping features from the sparse domain back to the high-dimensional feature space, yielding the final hyperspectral feature representation

$$F_H^i = \text{PWConv}\left(F_{sph}^i\right) \tag{6}$$

where $F_H^i \in \mathbb{R}^{P_i \times P_i \times B}$ represents the final output of the MS-SSSE module as a hyperspectral feature representation.

B. MS-GSSE Module

In order to address the "geometric sparsity" characteristic of LiDAR data, the MS-GSSE module was designed, as illustrated in Fig. 1(c). The module commences with an examination of geometric contours through the utilization of deep separable directional convolutions. Subsequently, an embedded sparse encoding layer is employed to enhance sparsity, thereby achieving native feature sparsity. The module accomplishes this process through three steps: 1) geometric feature extraction; 2) sparse solving; and 3) feature reconstruction.

1) *Geometric Feature Extraction:*

The single-channel LiDAR high-dimensional data are mapped to a B-dimensional feature space via a 1×1 convolution

$$F_l^i = \text{Conv}_{1\times1}\left(X_L^i\right) \tag{7}$$

where $F_l^i \in \mathbb{R}^{P_i \times P_i \times B}$ are the projected B-dimensional features.

In order to efficiently capture directional geometric features across multiple scales in LiDAR data, a depth-separable directional convolutional architecture with three scales was designed. In the context of the scale (k_1^i, k_2^i), the values of $i \in \{s, m, l\}$ are determined as follows: $k_1^i \in \{3, 5, 7\}$ and $k_2^i \in \{5, 9, 13\}$, where $k_2^i > k_1^i$.

Horizontal Direction: The employment of depthwise separable convolution with a kernel size of (k_1^i, k_2^i) is recommended, with the objective being the primary capture of spatial features in the horizontal direction

$$F_{lh}^i = \text{PW}_{k_1^i \times k_1^i}\left(\text{DW}_{k_1^i \times k_2^i}\left(F_f^i\right)\right) \tag{8}$$

where $F_{lh}^i \in \mathbb{R}^{P_i \times P_i \times B}$ is the horizontal direction feature, DW denotes depthwise convolution, and PW denotes pointwise convolution.

Vertical Direction: The use of depthwise separable convolution with a kernel size of (k_2^i, k_1^i) is recommended for the purpose of extracting spatial features in the vertical direction

$$F_{lv}^i = \text{PW}_{k_2^i \times k_2^i}\left(\text{DW}_{k_2^i \times k_1^i}\left(F_f^i\right)\right) \tag{9}$$

where $F_{lv}^i \in \mathbb{R}^{P_i \times P_i \times B}$ is the vertical direction feature, DW denotes depthwise convolution, and PW denotes pointwise convolution. These two directional strip convolutions enhance edge and contour features in the horizontal and vertical directions, respectively, through asymmetric rectangular convolution kernels. The horizontal and vertical features are then concatenated

$$F_{ls}^i = \text{Conv}_{1\times1}\left(\text{Concat}\left(F_{lh}^i, F_{lv}^i\right)\right) \tag{10}$$

where $F_{ls}^i \in \mathbb{R}^{P_i \times P_i \times B}$ are the concatenated multidirectional geometric features.

2) *Sparse Solving:*

In this step, $S_{\lambda_l}(\cdot)$ is also introduced to enforce its sparsity

$$F_{spl}^i = S_{\lambda_l}\left(F_{ls}^i\right) \triangleq \text{sign}\left(F_{ls}^i\right) \cdot \max\left(|F_{ls}^i| - \lambda_l, 0\right) \tag{11}$$

where F_{spl}^i are the sparsified geometric features and λ_l is a learnable geometric sparsity threshold. The operation in

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question is designed to preserve the most responsive edges and contour features while suppressing the smooth response in homogeneous regions.

3) *Feature Reconstruction:*

Sparse geometric features F^i_{spl} undergo channel fusion and feature reconstruction through a 1×1 convolution, yielding the final LiDAR feature representation

$$F^i_L = \mathrm{Conv}_{1\times1}\left(F^i_{spl}\right) \tag{12}$$

where $F^i_L \in \mathbb{R}^{P_i \times P_i \times B}$ denotes the ultimate geometric feature representation generated by the MS-GSSE module.

C. CSS Module

As demonstrated in Fig. 1(a), within the CSS-Mamba framework, the CSS module employs VSS blocks to capture long-range spatial dependencies within features across diverse scales. Subsequently, the hierarchical application of SCS blocks establishes explicit competition between multiscale features and completes sparse feature aggregation. Unlike conventional multiscale fusion strategies such as weighted averaging or feature concatenation, which treat all scale features indiscriminately and tend to retain redundant responses across scales, the CSS module introduces an explicit competition mechanism between adjacent-scale features. Specifically, for each spatial position, the competition weights assigned to two scales are normalized via Softmax, ensuring that their sum equals one. This complementarity constraint forces the model to make an adaptive selection between competing scales rather than simply aggregating them, thereby suppressing redundant scale responses while amplifying discriminative ones.

1) *VSS Block:*

The fundamental principle of the VSS block [47] is predicated on the selective scan 2-D (SS2D) method, as illustrated in Fig. 2(b). This technique unfolds image blocks along four distinct directions: top-left to bottom-right, bottom-right to top-left, top-right to bottom-left, and bottom-left to top-right, generating four separate sequences. Consequently, each feature sequence (scan) is subjected to processing by the S6 block. Finally, these sequences are reconfigured back into a single image. Given the input feature z, the output feature $\bar{z}$ of SS2D can be expressed as

$$z_v = \mathrm{expand}\,(z, v) \tag{13}$$

$$\bar{z}_v = \mathrm{S6}\,(z_v) \tag{14}$$

$$\bar{z} = \mathrm{merge}\,(\bar{z}_1, \bar{z}_2, \bar{z}_3, \bar{z}_4) \tag{15}$$

where $v \in V = \{1, 2, 3, 4\}$ represents four distinct scanning directions. The operators Expand $(\cdot)$ and Merge $(\cdot)$ are analogous to scan expansion and scan merging operations, respectively. The VSS block architecture is illustrated in Fig. 2(a). The module under consideration draws inspiration from the Transformer block design and consists of two serially connected sublayers, each incorporating residual connections. The input features initially enter the first sublayer and propagate along two paths: one serves as the residual path, while the main processing path sequentially feeds the input features into the LN and SS2D block modules. The SS2D module integrates a series of operations, including linear projection,

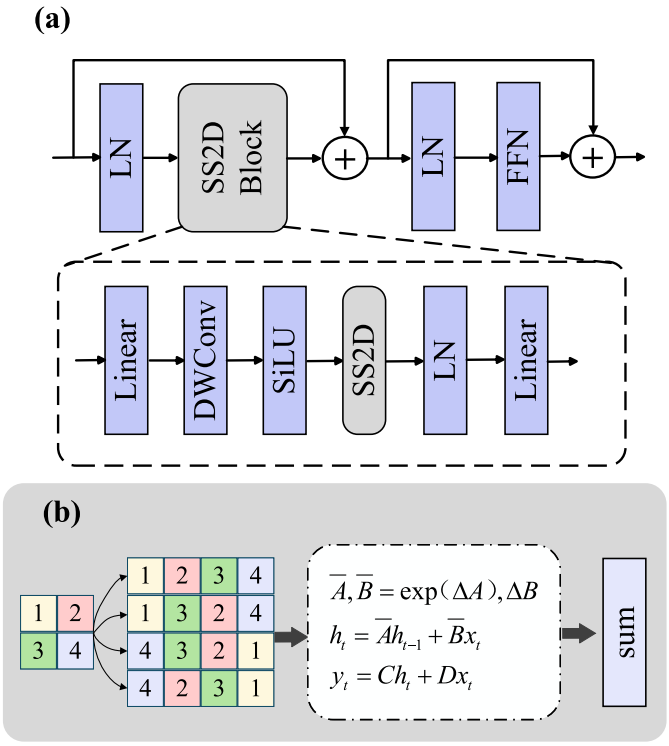


Fig. 2. VSS block architecture. (a) VSS block. (b) SS2D.

depthwise separable convolution, SiLU activation, and the core SS2D operation. In conclusion, the primary processing path's output is appended to the original input feature transmitted via the residual path, thereby concluding the initial sublayer's computation. The output then serves as input for the second sublayer. As with the preceding structure, features are processed through both residual and main processing paths. Within the primary processing pathway, features undergo layer normalization prior to being introduced into a standard feed-forward network (FFN) for nonlinear feature transformation. Finally, the FFN output is elementwise added to the sublayer's input (i.e., the output of the first sublayer), forming the final output of the VSS block.

2) *SCS Block:*

The SCS block constitutes the core component of CSS, designed to perform SCS between any two features at different scales.

Given input features $A \in \mathbb{R}^{C\times H\times W}$ and $B \in \mathbb{R}^{C\times H\times W}$ (which represent features at two different scales), the SCS module receives the fused feature $U \in \mathbb{R}^{C\times H\times W}$ (passed in by CSS after adding the two scale features). The process commences with a reduction of the number of channels to 2 via a 1×1 convolution. Subsequently, a Softmax function is employed to generate pixelwise competition weights $W \in \mathbb{R}^{2\times H\times W}$ in the channel dimension

$$W = \mathrm{Softmax}\left(\mathrm{Conv}_{1\times1}\left(U\right)\right). \tag{16}$$

It is important to note that Softmax is employed along the channel dimension, thereby ensuring that for each spatial position (h, w), the weight values of the two channels satisfy

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the complementarity constraint. Subsequently, the weight map W is decomposed into two weight components: $W_A \in \mathbb{R}^{1\times H\times W}$ and $W_B \in \mathbb{R}^{1\times H\times W}$, corresponding to the competitive weights at the two scales. The weights are applied to the original features through elementwise multiplication, with residual connections added to preserve the original feature information

$$\begin{aligned} A' &= W_A \odot A + A \\ B' &= W_B \odot B + B \end{aligned} \tag{17}$$

where (A', B') denotes the feature pair weighted by competition, which is the output of the SCS block.

3) *Hierarchical Application and Dynamic Fusion of CSS Module:*

To establish a comprehensive competitive relationship across multiple scales (large, medium, and small), CSS modules employ a hierarchical application strategy for SCS submodules. During the input phase of CSS modules, features X_L, X_M, and X_S (corresponding to large, medium, and small scales, respectively) are initially enhanced through the VSS block and batch normalization (BN) layer to model intrascale long-range dependencies, yielding scaled features X_{L1}, X_{M1}, X_{S1}

$$\begin{aligned} X_{L1} &= \mathrm{BN}\left(\mathrm{VSS}\left(X_L\right)\right) \\ X_{M1} &= \mathrm{BN}\left(\mathrm{VSS}\left(X_M\right)\right) \\ X_{S1} &= \mathrm{BN}\left(\mathrm{VSS}\left(X_S\right)\right). \end{aligned} \tag{18}$$

Consequently, the CSS module employs SCS blocks in parallel across two hierarchical levels to execute competitive fusion. The design is predicated on the principle of "adjacent-scale competition," which is to be distinguished from direct competition between scales that are too distant from each other. The purpose of this is to enhance the stability and rationality of the fusion process.

Level I Medium- to Large-Scale Competition: Apply SCS blocks to perform SCS on large-scale enhanced features X_{L1} and medium-scale enhanced features X_{M1} for SCS

$$\left(X_{LM}, X_{ML}\right) = \mathrm{SCS}\left(X_{L1}, X_{M1}\right) \tag{19}$$

where (X_{LM}, X_{ML}) are the feature pair after competition weighting for the large and medium scales.

Level II Small- to Medium-Scale Competition: Apply SCS blocks to perform spatial competitive fusion on medium-scale enhanced features X_{S1} and small-scale enhanced featuresX_{M1} for SCS

$$\left(X_{MS}, X_{SM}\right) = \mathrm{SCS}\left(X_{M1}, X_{S1}\right) \tag{20}$$

where (X_{MS}, X_{SM}) is the feature pair after competition weighting for the medium and small scales.

Subsequently, elementwise multiplication is performed for each weighted feature within each competitive layer. This process enhances the discriminative influence of the competitive mechanism on the features

$$\begin{aligned} Y_1 &= X_{ML} \odot X_{LM} \\ Y_2 &= X_{SM} \odot X_{MS} \end{aligned} \tag{21}$$

where Y_1 and Y_2 are the outputs of the two competition levels, respectively. This feature multiplication operation effectively suppresses feature responses in regions with lower competitive weights while amplifying them in regions with higher weights, thereby achieving stronger competitive discrimination. Concurrently, a learnable fusion weight is introduced with the aim of establishing a global dynamic equilibrium between the two competing groups. This weight is learned through GAP and a lightweight multilayer perceptron (MLP)

$$\alpha = \mathrm{Sigmoid}\left(\mathrm{MLP}\left(\mathrm{GAP}\left(Y_1\right) + \mathrm{GAP}\left(Y_2\right)\right)\right). \tag{22}$$

The Sigmoid function compresses the output of the MLP into the range (0, 1), resulting in the learnable fusion weight α. The final CSS module outputs features obtained through weighted fusion of the two competing outputs from these layers

$$Y_{CSS} = \alpha \cdot Y_1 + (1-\alpha)\cdot Y_2 \tag{23}$$

where $Y_{CSS} \in \mathbb{R}^{P\times P\times B}$ denotes the final output of the CSS module, which incorporates competitive discriminative information from multiscale features.

IV. Experiment

The objective of this section is to methodically assess the performance, efficiency, and generalization capability of the proposed CSS-Mamba model in the context of multimodal remote sensing image classification tasks. Considering the variations in spectral, spatial, and elevation information distribution across different remote sensing scenarios, three publicly available benchmark datasets encompassing agricultural landscapes, complex urban environments, and multiclass mixed areas were selected to ensure the representativeness and broad applicability of the experimental results. In the experimental design, we not only compared CSS-Mamba with multiple state-of-the-art multimodal methods to comprehensively characterize its advantages under different fusion strategies but also conducted in-depth analyses from multiple dimensions to dissect the model's effectiveness, robustness, and efficiency through ablation studies, statistical significance tests, parameter sensitivity analysis, t-distributed stochastic neighbor embedding (t-SNE) feature visualization, sparsity quantification, computational complexity analysis, and experiments with varying training sample proportions. The comparative results are reported as mean $\pm$ standard deviation over five independent runs. Sections IV-A–IV-I will systematically present and discuss the experimental results concerning these aspects.

A. Datasets

Houston 2013: The Houston 2013 dataset, provided by NASA, encompasses an urban area in Houston, TX, comprising 144 hyperspectral bands (0.38–1.05 μm) and a single-band LiDAR DSM. The spatial resolution of the image is 2.5 m, and its dimensions are 349×1905 pixels. The image encompasses 15 distinct urban classes, including buildings, roads, bare soil, and various vegetation types. The presence of strong heterogeneity and shadow effects has been demonstrated to result in an increase in the complexity of spectral–spatial coupling. As illustrated in Fig. 3, the pseudocolor composite, DSM, and ground truth are presented.

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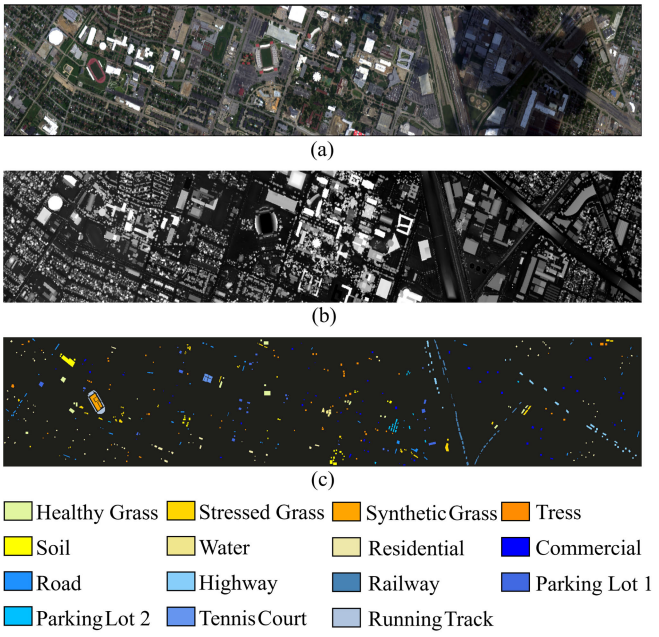


Fig. 3. Houston 2013 dataset. (a) Pseudocolor composite image based on bands, (b) LiDAR-derived DSM, and (c) ground-truth map with 15 heterogeneous urban classes.

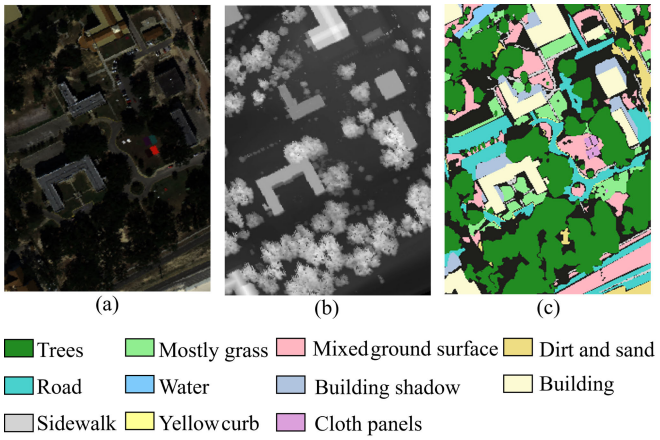


Fig. 4. MUUFL dataset. (a) Pseudocolor composite image based on bands, (b) LiDAR-derived DSM, and (c) ground-truth map with 11 imbalanced land-cover classes.

MUUFL: The MUUFL dataset, which was collected over the University of Southern Mississippi (Long Beach, MS, USA), comprises 64 hyperspectral bands (0.38–1.05 μm) and a LiDAR DSM at 1064 nm. The spatial resolution of the image is approximately 1 m, with an image size of 325×220 pixels, covering 11 classes. The presence of large homogeneous regions and severely underrepresented classes (e.g., Yellow Curb and Cloth Panels) renders this dataset challenging in the context of class imbalance. Fig. 4 presents a visual representation of the pseudocolor composite, DSM, and the ground truth.

Trento: The Trento dataset, which was captured in an agricultural landscape near Trento, Italy, comprises 63 hyperspectral bands (0.42–0.99 μm) and a LiDAR-derived digital surface model (DSM). The spatial resolution of the image is

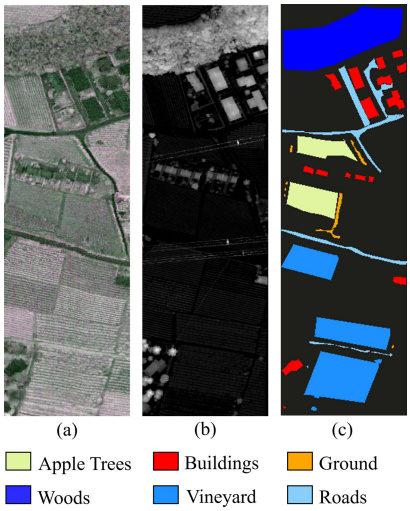


Fig. 5. Trento dataset. (a) Pseudocolor composite image based on bands, (b) LiDAR-derived (DSM) in gray scale, and (c) ground-truth map with six spectrally similar land-cover classes.

1 m, with an image size of 166×600 pixels that encompasses six land-cover classes. The acquisition of LiDAR data was undertaken utilizing an ALTM 3100EA sensor. It is evident that the presence of spectral similarity among vegetation types renders this scene particularly challenging from a fine-grained discrimination perspective. Fig. 5 presents a pseudocolor composite, DSM, and the ground-truth map.

The configuration of training and testing data for the three datasets is presented in Table I.

B. Experimental Setup

In order to validate the advanced nature and effectiveness of the proposed CSS-Mamba model, this study selected eight multimodal joint classification methods based on HSI and LiDAR for comparison. The aforementioned methods encompass a variety of mainstream architectures and fusion strategies, including, but not limited to, graph convolutional networks, Transformers, and Mamba. The following methods are utilized as comparative benchmarks: the extended vision Transformer (ExViT) [35], the hierarchical CNN and Transformer (HCT) [37], the graph-attention-based multimodal fusion (GAMF) [48], the Cross Hyperspectral (Cross-HL) and LiDAR Attention Transformer [36], the multiple information collaborative fusion network (MICF-Net) [20], the multiscale spatial–spectral cross-modal attention network (MS2CANet) [49], multiscale feature fusion Mamba (MSFMamba) [15], and the dual-feature attention hybrid graph convolutional network Mamba network (DAHGMN) [45]. All experiments were implemented using PyTorch on a hardware environment comprising 16 vCPUs, Intel[1] Xeon[1] Platinum 8358P CPU at 2.60 GHz paired with an RTX 4090 (24 GB). The Adam optimizer was employed with an initial learning rate of 5×10^{-5}, a batch size of 64, and 200 training epochs. In instances where the original parameters for comparison methods were

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TABLE I

LAND-COVER CLASSES AND THE CORRESPONDING NUMBERS OF TRAINING AND TESTING SAMPLES

Houston 2013 dataset			**MUUFL dataset**			**Trento dataset**		
Class Name	Training	Test	Class Name	Training	Test	Class Name	Training	Test
Healthy Grass	20	1231	Trees	20	23226	Apple Trees	20	4014
Stressed Grass	20	1234	Mostly Grass	20	4250	Buildings	20	2883
Synthetic Grass	20	677	Mixed Ground Surface	20	6862	Ground	20	459
Trees	20	1224	Dirt and Sand	20	1806	Woods	20	9103
Soil	20	1222	Road	20	6667	Vineyard	20	10481
Water	20	305	Water	20	446	Roads	20	3154
Residential	20	1248	Buildings Shadow	20	2213			
Commercial	20	1224	Buildings	20	6220			
Road	20	1232	Sidewalk	20	1365			
Highway	20	1207	Yellow Curb	20	163			
Railway	20	1215	Cloth Panels	20	249			
Parking Lot1	20	1213						
Parking Lot2	20	449						
Tennis Court	20	408						
Running Track	20	640						
Total	**120**	**30094**	**Total**	**220**	**53467**	**Total**	**300**	**14729**

TABLE II

QUANTITATIVE CLASSIFICATION RESULTS ON THE HOUSTON 2013 DATASET (MEAN ± STD OVER FIVE INDEPENDENT RUNS). THE BEST MEAN IN EACH ROW IS HIGHLIGHTED IN BOLD

NO.	Class Name	Houston 2013 dataset								
		ExViT	HCT	GAMF	Cross-HL	MICF-Net	MS2CANet	MSFMamba	DAHGMN	CSS-Mamba
C1	Healthy Grass	89.96±8.46	96.02±1.30	**100.00±00.00**	77.56±2.15	90.61±5.01	93.11±2.92	92.15±0.71	92.46±3.24	92.06±3.09
C2	Stressed Grass	94.38±4.24	95.36±3.44	79.88±4.18	81.42±4.35	93.48±4.81	89.90±3.31	95.66±1.96	90.36±4.77	**97.33±1.38**
C3	Synthetic Grass	99.26±0.37	99.38±0.59	99.25±0.11	94.45±2.18	99.58±0.40	99.11±0.98	99.35±0.12	98.94±1.14	**99.82±0.18**
C4	Trees	94.90±4.80	99.23±0.59	81.44±1.08	92.24±6.14	96.21±2.38	**99.35±0.72**	91.31±1.54	98.50±0.78	98.92±1.59
C5	Soil	94.34±7.83	99.90±0.13	99.53±0.39	92.76±1.10	97.71±1.72	99.74±0.27	99.82±0.14	99.72±0.39	**99.92±0.13**
C6	Water	94.30±5.09	99.02±1.65	78.84±3.93	89.12±2.51	95.93±1.53	97.51±2.79	97.31±1.73	96.59±3.47	**99.38±0.91**
C7	Residential	91.92±3.80	91.06±3.91	81.21±0.47	87.40±7.25	**95.97±2.05**	94.36±2.51	90.40±4.82	95.75±2.12	94.37±1.98
C8	Commercial	87.08±11.68	81.06±4.33	71.99±3.63	81.28±1.33	83.93±4.17	86.46±2.58	91.45±0.84	85.39±5.66	**93.42±1.62**
C9	Road	86.09±11.79	78.99±6.22	51.01±3.92	64.94±3.94	81.84±4.51	82.95±2.80	79.66±1.60	90.67±5.66	**90.86±2.06**
C10	Highway	78.96±9.23	**95.46±3.42**	78.11±2.06	88.92±2.35	76.76±7.39	90.46±3.50	94.37±2.15	79.24±5.30	94.16±1.69
C11	Railway	93.98±3.93	97.12±2.00	92.62±4.33	95.57±2.04	88.43±4.89	95.01±4.10	**97.27±0.47**	97.17±2.24	93.75±1.56
C12	Parking Lot1	78.42±12.07	83.23±4.82	41.88±1.94	63.35±7.58	83.21±7.97	93.11±0.58	87.83±1.28	85.94±13.72	**93.92±3.31**
C13	Parking Lot2	95.55±2.47	**97.59±1.93**	89.49±1.85	84.54±4.64	89.25±3.45	97.10±1.78	94.83±0.38	95.81±2.18	95.58±1.85
C14	Tennis Court	99.66±0.37	**100.00±0.00**	**100.00±00.00**	99.12±0.36	99.90±0.20	99.26±1.47	99.71±0.59	**100.00±00.00**	99.95±0.10
C15	Running Track	98.03±3.49	**100.00±0.00**	**100.00±00.00**	89.96±2.90	99.97±0.06	98.87±2.10	**100.00±00.00**	99.81±0.23	**100.00±0.00**
	OA(%)	90.49±1.31	93.01±1.08	80.73±2.08	83.74±1.93	90.28±2.06	93.47±0.97	93.08±0.75	92.71±1.01	**95.57±0.38**
	AA(%)	91.79±1.20	94.23±0.87	83.02±1.99	85.51±1.68	91.52±1.72	94.42±0.82	94.07±0.70	93.76±0.88	**96.23±0.35**
	Kappa(%)	89.72±1.42	92.43±1.16	79.16±2.25	82.38±2.09	89.49±2.23	92.94±1.04	92.51±0.81	92.12±1.09	**95.20±0.42**

not available, we undertook parameter tuning under identical training strategies to ensure fairness.

C. Comparisons With Previous Methods

Quantitative Performance: Tables II–IV present the quantitative analysis results of different methods across three datasets. On the Houston 2013 dataset, CSS-Mamba achieves an OA of 95.57%± 0.38%, surpassing MSFMamba (93.08%± 0.75%), MS2CANet (93.47% ± 0.97%), and DAHGMN (92.71% ± 1.01%). The dataset comprises 15 highly heterogeneous urban categories, presenting considerable challenges to scale adaptability and feature discriminative power. CSS-Mamba demonstrates particular advantages in spectrally and spatially confusable categories. In C9 Road, CSS-Mamba attains 90.86%± 2.06%, surpassing GAMF (51.01%± 3.92%) by 39.85 percentage points and Cross-HL (64.94% ± 3.94%) by 25.92 percentage points, while maintaining substantially lower variance than DAHGMN (90.67% ± 5.66%). In C10 Highway and C11 Railway, CSS-Mamba achieves 94.16% ± 1.69% and 93.75% ± 1.56%, respectively, outperforming the majority of comparison methods. In C8 Commercial and C12 Parking Lot1, CSS-Mamba attains 93.42% ± 1.62% and 93.92% ± 3.31%, compared to HCT (81.06% ± 4.33% and 83.23% ± 4.82%) and GAMF (71.99% ± 3.63% and 41.88%± 1.94%). Notably, CSS-Mamba maintains the lowest OA standard deviation (0.38%) among all methods, demonstrating consistent discriminative performance across different random initializations.

On the MUUFL dataset, CSS-Mamba achieves an OA of 86.21% ± 0.99%, surpassing DAHGMN (80.94% ± 1.36%) by 5.27 percentage points, MS2CANet (81.05%± 1.28%) by 5.16 percentage points, and MSFMamba (75.77%± 0.81%) by 10.44 percentage points. The dataset is characterized by severe

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TABLE III

QUANTITATIVE CLASSIFICATION RESULTS ON THE MUUFL DATASET (MEAN ± STD OVER FIVE INDEPENDENT RUNS). THE BEST MEAN IN EACH ROW IS HIGHLIGHTED IN BOLD

NO.	Class Name	MUUFL dataset								
		ExViT	HCT	GAMF	Cross-HL	MICF-Net	MS2CANet	MSFMamba	DAHGMN	CSS-Mamba
C1	Trees	87.29±4.49	84.18±1.05	87.40±2.58	83.55±2.42	81.68±4.88	89.24±2.00	78.89±0.86	91.17±2.15	**92.20±0.25**
C2	Mostly Grass	55.74±9.88	74.48±7.48	67.01±3.40	75.86±3.37	69.40±3.39	77.00±1.71	68.61±1.90	72.00±4.24	**84.16±0.48**
C3	Mixed Ground Surface	69.19±5.21	55.49±4.92	**79.94±0.22**	55.77±3.99	56.67±5.69	60.55±5.93	60.89±2.34	57.84±8.65	68.74±1.10
C4	Dirt and Sand	81.92±12.76	90.94±4.38	87.07±4.65	85.20±5.38	75.51±7.32	90.24±3.65	89.83±1.75	78.68±8.86	**93.08±2.33**
C5	Road	76.87±4.20	72.20±10.12	67.24±2.27	81.01±2.50	76.78±2.74	72.88±4.47	72.38±2.24	69.76±6.98	**81.92±0.72**
C6	Water	99.60±0.41	98.83±1.20	80.50±4.30	**100.00±00.00**	95.03±3.74	99.34±1.07	**100.00±00.00**	98.07±1.97	99.88±0.05
C7	Buildings Shadow	85.99±5.10	83.41±6.81	81.96±0.18	82.73±1.87	74.89±12.55	83.54±6.02	72.70±5.47	81.52±6.62	**87.63±4.23**
C8	Buildings	92.98±1.32	88.31±3.43	86.84±3.34	85.63±6.11	63.47±13.37	83.82±8.17	90.36±0.87	92.37±1.94	**93.19±1.28**
C9	Sidewalk	43.46±6.49	44.91±4.81	47.10±6.09	**63.21±5.48**	56.07±5.12	59.92±6.66	42.93±1.37	48.97±17.96	51.46±7.83
C10	Yellow Curb	65.52±7.87	**85.28±5.85**	59.60±3.04	57.34±2.61	76.25±8.16	83.09±6.98	83.44±1.55	72.15±5.56	70.01±1.26
C11	Cloth Panels	98.07±1.71	97.67±1.96	98.42±1.14	89.72±2.59	97.03±1.73	96.31±3.52	**99.44±0.20**	91.57±7.02	98.15±0.26
	OA(%)	80.55±1.89	78.10±1.86	80.89±2.16	78.89±3.27	73.84±3.34	81.05±1.28	75.77±0.81	80.94±1.36	**86.21±0.99**
	AA(%)	77.88±1.98	79.61±1.15	76.64±2.84	78.18±3.30	74.80±1.89	81.45±1.48	78.13±0.90	77.64±3.47	**83.67±1.80**
	Kappa(%)	74.97±2.16	72.10±2.35	74.71±3.26	72.07±4.33	66.65±4.00	75.46±1.56	69.41±1.01	75.25±1.84	**81.75±1.30**

TABLE IV

QUANTITATIVE CLASSIFICATION RESULTS ON THE TRENTO DATASET (MEAN ± STD OVER FIVE INDEPENDENT RUNS). THE BEST MEAN IN EACH ROW IS HIGHLIGHTED IN BOLD

NO.	Class Name	Trento dataset								
		ExViT	HCT	GAMF	Cross-HL	MICF-Net	MS2CANet	MSFMamba	DAHGMN	CSS-Mamba
C1	Apple Trees	97.87±1.21	98.83±0.57	97.15±0.96	94.85±2.07	98.31±0.74	99.64±0.21	93.46±3.55	98.63±1.02	**99.65±0.23**
C2	Buildings	98.00±1.14	96.24±2.59	94.60±2.79	92.30±1.67	97.76±1.89	97.80±0.82	97.73±0.69	**98.72±0.90**	98.42±1.01
C3	Ground	96.69±2.66	98.43±1.04	**99.38±0.20**	84.34±3.89	99.02±1.08	99.26±0.75	97.04±0.47	97.82±1.57	99.02±0.78
C4	Woods	**100.00±00.00**	**100.00±00.00**	97.99±0.21	99.65±0.34	**100.00±00.00**	**100.00±00.00**	99.83±0.14	**100.00±00.00**	**100.00±00.00**
C5	Vineyard	99.56±0.42	97.27±2.41	97.08±2.27	94.22±3.78	99.34±0.44	99.95±0.04	97.42±1.70	99.92±0.06	**99.96±0.01**
C6	Roads	95.34±3.25	93.94±1.88	94.30±0.58	86.78±5.16	96.51±2.74	97.00±1.64	92.55±1.20	**97.42±1.93**	97.30±1.78
	OA(%)	98.83±0.33	97.87±0.93	96.87±1.31	94.85±1.23	98.95±0.18	99.40±0.13	97.13±1.08	99.36±0.27	**99.49±0.22**
	AA(%)	97.91±0.73	97.45±0.60	96.75±1.17	92.02±1.27	98.49±0.37	98.94±0.31	96.34±0.94	98.75±0.53	**99.06±0.29**
	Kappa(%)	98.44±0.45	97.17±1.22	95.82±1.75	93.16±1.62	98.60±0.24	99.20±0.18	96.18±1.44	99.15±0.31	**99.32±0.23**

class imbalance and the coexistence of large homogeneous regions with small heterogeneous targets. In the dominant C1 Trees category, CSS-Mamba attains 92.20% ± 0.25%, while MSFMamba achieves only 78.89% ± 0.86%, a gap of 13.31 percentage points. In C2 Mostly Grass, CSS-Mamba achieves 84.16% ± 0.48%, surpassing DAHGMN (72.00% ± 4.24%) by 12.16 percentage points, with the latter's high variance indicating systematic instability in spectrally ambiguous transition regions. CSS-Mamba ranks first in eight out of 11 categories, demonstrating consistent superiority across both dominant and minority classes.

The results on the Trento dataset further validate the effectiveness of CSS-Mamba, which achieves an OA of 99.49% ± 0.22% and consistently outperforms all comparison methods despite the majority of methods attaining high accuracy (> 97%) on this dataset.

Statistical Significance Test: To rigorously assess whether the reported performance gains are statistically reliable, we conducted paired t-tests on the per-run OA values between CSS-Mamba and the second-best method (MS2CANet) on each benchmark dataset, based on five independent runs with different random seeds. CSS-Mamba achieves statistically significant improvements over MS2CANet on the Houston 2013 dataset ($p = 0.0013$) and the MUUFL dataset ($p < 0.0001$). On the Trento dataset, the result does not reach statistical significance ($p = 0.1012$) due to a ceiling effect inherent to near-saturated accuracy regimes; the advantage of CSS-Mamba is instead reflected in its consistent per-class superiority (see Table IV) and lower performance variance across runs.

Qualitative Analysis: As demonstrated in Figs. 6–8, a clear visual representation of the classification performance disparities among diverse methodologies is provided. The visual classification results generated by the proposed CSS-Mamba method demonstrate higher consistency with ground-truth maps, showcasing superior classification performance.

In the Houston 2013 dataset, the boundary accuracy in complex urban scenes emerges as the key factor distinguishing the performance of the methods. As demonstrated in Fig. 6, significant boundary blurring is exhibited by GAMF and MSFMamba at the intersection of "C9, Road" and "C8, Residential," where road edges are misclassified as residential areas or other categories, resulting in distorted road geometry. The CSS-Mamba method is an innovative approach that enhances discriminative features in small-scale regions through a competitive selection mechanism. This mechanism generates precise category segmentation boundaries that preserve the linear continuity of roads.

In the MUUFL dataset, the capacity to manage boundaries between buildings and shadows directly correlates with a

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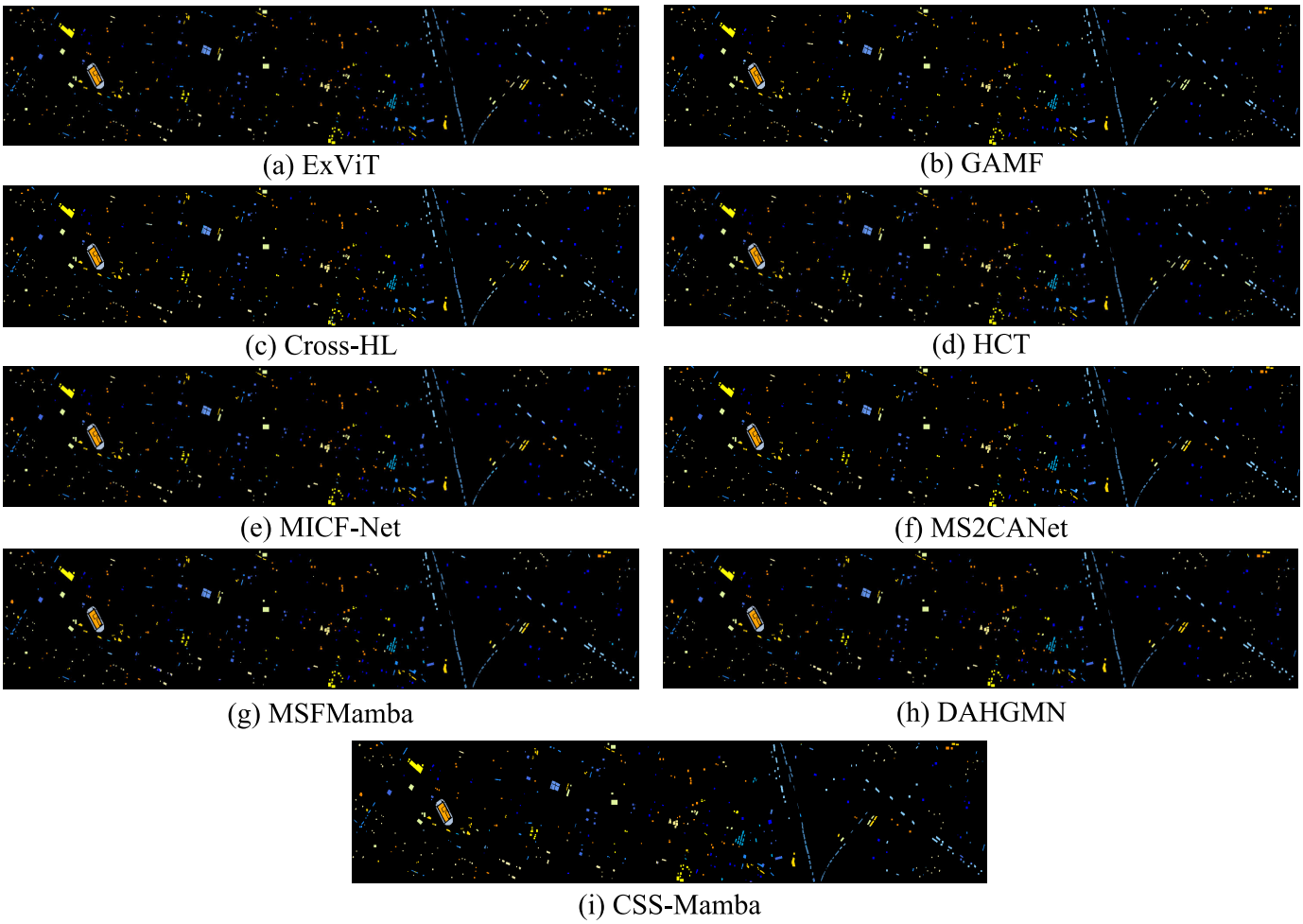


Fig. 6. Classification maps of the Houston 2013 data generated by different methods. (a) ExViT (94.09%). (b) HCT (94.23%). (c) GAMF (80.83%). (d) Cross-HL (88.26%). (e) MICF-net (90.36%). (f) MS2CANet (93.21%). (g) MSFMamba (94.26%). (h) DAHGMN (94.61%). (i) CSS-Mamba (95.29%). Overall accuracies are reported in parentheses.

method's robustness. The majority of comparison methods demonstrate a substantial degree of boundary confusion when processing the "C7, Buildings Shadow" and "C8, Buildings" categories. The transition zone between these two classes displays salt-and-pepper noise and irregular, jagged boundaries, resulting in poor spatial consistency in the classification map. In contrast, the CSS-Mamba method not only accurately identified these two highly similar categories but also generated clearer and smoother classification boundaries along the edges of building shadows.

In the Trento dataset, the discriminative capability for spectrally similar categories became a key metric for evaluating method performance. The majority of comparison methods (e.g., ExViT, Cross-HL, and GAMF) misclassified pixels from part of the "C5, Vineyard" category as "C1, Apple Trees," indicating significant confusion in the boundary region between the two classes. Conversely, the CSS-Mamba method has been shown to almost entirely circumvent such misclassifications, thereby ensuring the delineation of distinct category boundaries at orchard boundaries. This exceptional performance is closely aligned with the quantitative results presented in Table IV for C1 (99.65% $\pm$ 0.23%) and C5 (99.96% $\pm$ 00.01%).

D. Ablation Studies

1) *Ablation on Key Components:*

In the context of ablation experiments, the contribution of each constituent element, including MS-SSSE, MS-GSSE, and CSS, is evaluated using OA, AA, and the Kappa coefficient. Table V presents detailed experimental results across three benchmark datasets. The baseline model employs a conventional dual-branch convolutional architecture coupled with a fully connected classifier, representing the current standard processing paradigm for multisource data fusion.

First, the introduction of the MS-SSSE module has been demonstrated to enhance performance significantly. On the Houston 2013 dataset, the OA metric improved from the baseline of 79.25%–83.78%, representing an increase of 4.53 percentage points; gains of 4.43% and 3.08% were achieved on the MUUFL and Trento datasets, respectively. This enhancement serves to validate our fundamental hypothesis: namely, by extracting and utilizing spectral sparsity features from HSI data, we can effectively enhance the discriminative power of features. This module has been found to demonstrate particular strength on the Houston 2013 dataset, which exhibits high spectral complexity. Second, the synergistic effect between

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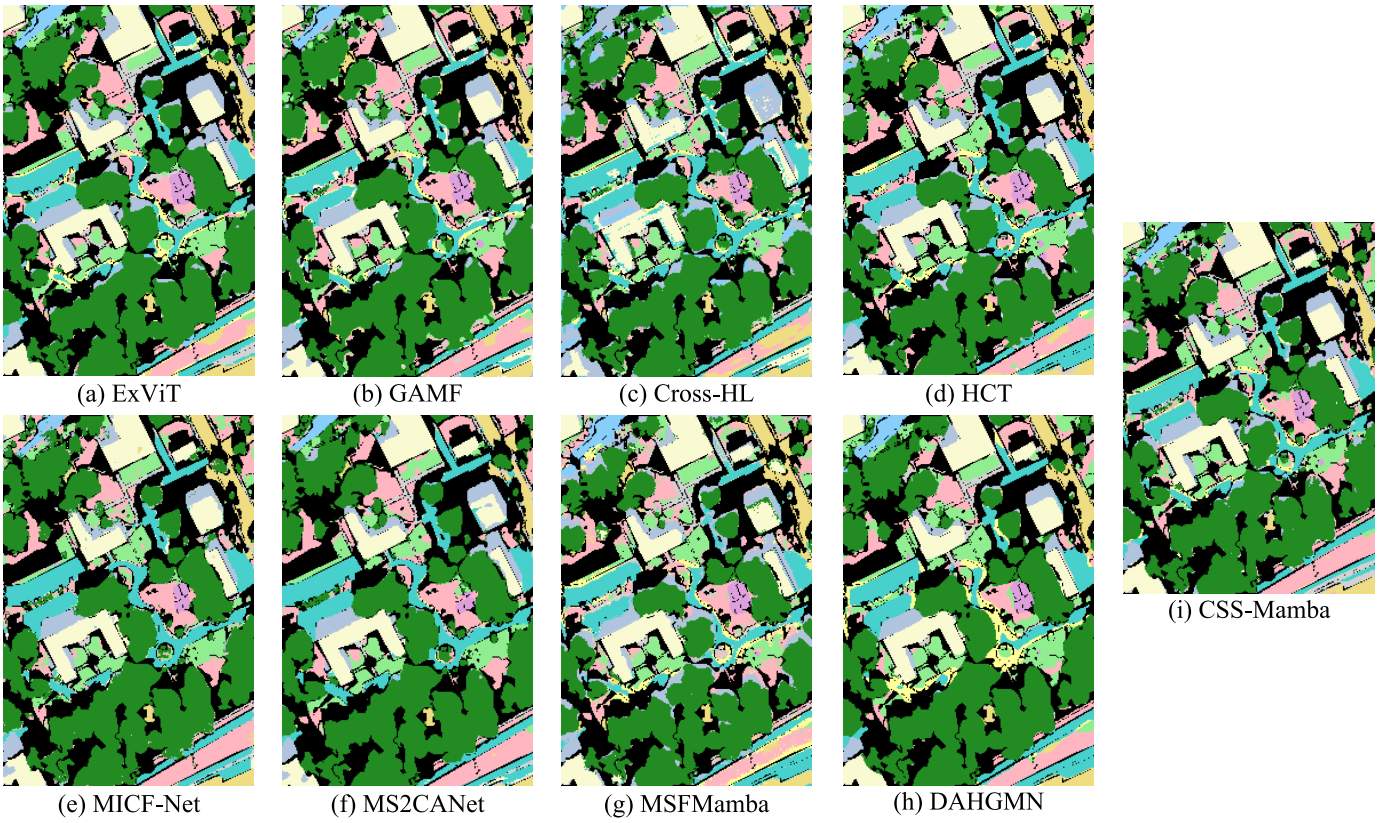


Fig. 7. Classification maps of the MUUFL data generated by different methods. (a) ExViT (82.36%). (b) HCT (81.67%). (c) GAMF (82.70%). (d) Cross-HL (77.24%). (e) MICF-net (79.29%). (f) MS2CANet (79.62%). (g) MSFMamba (76.73%). (h) DAHGMN (83.07%). (i) CSS-Mamba (86.80%). Overall accuracies are reported in parentheses.

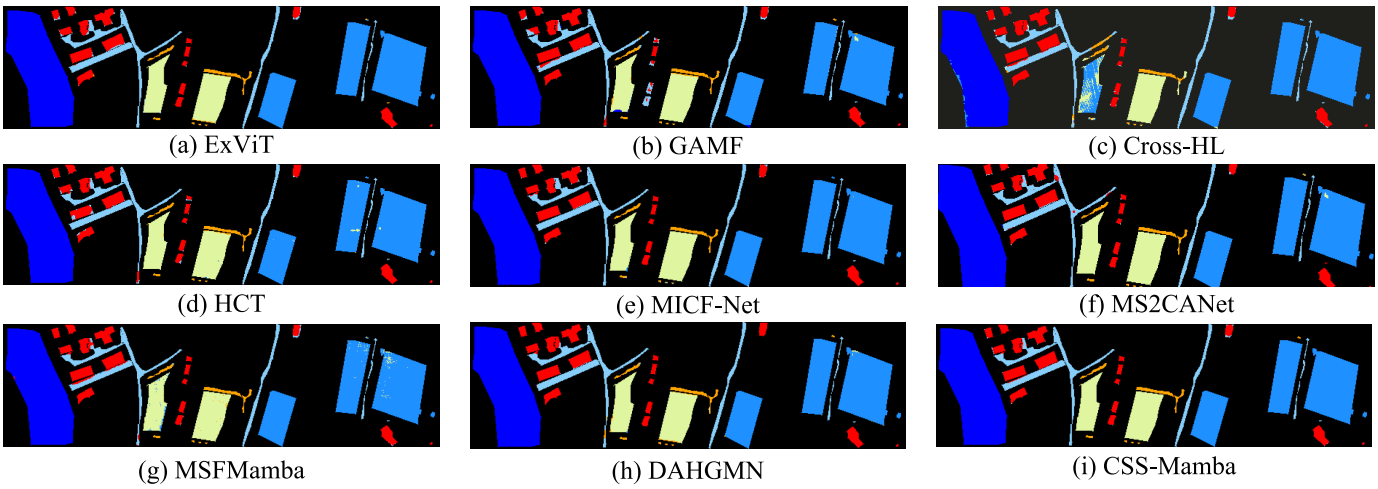


Fig. 8. Classification maps of the Trento data generated by different methods. (a) ExViT (99.28%). (b) HCT (98.10%). (c) GAMF (97.62%). (d) Cross-HL (94.19%). (e) MICF-net (99.17%). (f) MS2CANet (98.50%). (g) MSFMamba (98.30%). (h) DAHGMN (99.14%). (i) CSS-Mamba (99.60%). Overall accuracies are reported in parentheses.

the MS-GSSE module and the MS-SSSE module results in a further enhancement in performance. In comparison with the utilization of MS-SSSE in isolation, this module yielded supplementary accuracy enhancements of 5.47%, 3.63%, and 2.79% for Houston 2013, MUUFL, and Trento, respectively. This outcome emphasizes the importance of designing bespoke extraction strategies that are tailored to the geometric properties of LiDAR data. Orthogonal strip convolutions effectively capture structural information across different orientations, thereby supplementing the 3-D geometric information that spectral features alone are unable to express. With regard to the question of overall improvement, the complete architecture achieved substantial gains of 16.04%, 12.46%, and 7.59% over the baseline across the three datasets. The CSS module employs a CSS mechanism to dynamically select optimal scale features, thereby avoiding redundant stacking of multiscale features and adaptively enhancing key scale information according to the characteristics of different objects. It is

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TABLE V

ABLATION STUDY OF THE PROPOSED METHOD ACROSS THREE BENCHMARK DATASETS. RESULTS ARE COMPARED AMONG BASELINE, BASELINE + MS-GSSE, BASELINE + MS-SSSE + MS-GSSE, AND BASELINE + MS-SSSE' + MS-GSSE + CSS

Method	Houston 2013			MUUFL			Trento		
	OA(%)	AA(%)	Kappa(%)	OA(%)	AA(%)	Kappa(%)	OA(%)	AA(%)	Kappa(%)
Baseline	79.25	76.63	77.89	74.34	72.80	71.12	92.01	90.65	91.12
+ MS-GSSE	83.78	82.94	82.65	78.77	76.65	77.31	95.09	92.77	94.61
+ MS-SSSE + MS-GSSE	89.25	91.28	88.47	82.40	81.98	80.65	97.88	96.93	97.52
+ MS-SSSE + MS-GSSE + CSS	**95.29**	**96.03**	**94.90**	**86.80**	**85.05**	**82.83**	**99.60**	**99.34**	**99.46**

TABLE VI

ABLATION ANALYSIS OF DIFFERENT FUSION STRATEGIES ON THREE DATASETS

Fusion method	Houston 2013 OA(%)	MUUFL OA(%)	Trento OA(%)
Add	93.14	83.69	99.07
Concat+Conv	93.45	84.97	99.16
SE-based	94.05	85.47	99.21
SKNet-style	94.38	85.89	99.35
SCS Block	**95.29**	**86.80**	**99.60**

noteworthy that the observed variation in improvement across datasets is indicative of the adaptability of the module design to the characteristics of the data. The Houston 2013 dataset, representing an urban environment with complex man-made objects, exhibits the most pronounced multiscale properties and, thus, benefits most substantially from our multiscale architecture. The Trento dataset, despite its inherently high baseline accuracy, still achieved stable improvements, demonstrating the method's generalizability.

2) *Ablation Analysis of Fusion Strategies in CSS Module:*

To verify the effectiveness of the SCS mechanism in the SCS block, we replace it with four representative multiscale fusion strategies and report results in Table VI. Overall, the SCS block achieves the best classification accuracy across all three datasets, confirming the advantage of pixelwise spatial adaptive competition in multiscale feature selection. Add and Concat+Conv yield the lowest performance, as both lack explicit suppression of redundant scale responses across scales. SE-based fusion introduces channel attention but shares identical weights across all spatial positions, limiting its ability to model local heterogeneity at complex boundaries. SKNet-style fusion enables multibranch soft selection and performs closest to the proposed SCS block, yet its globally channelwise weights lack complementarity constraints, leaving residual scale redundancy. In contrast, the SCS block generates pixelwise competition weights with a Softmax complementarity constraint, enabling spatially adaptive scale selection and achieving the best performance across all three datasets.

E. Parameter Sensitivity Analysis

1) *Size of Input Patch:*

The selection of patch size is crucial for CSS-Mamba to fully exploit the multiscale characteristics of landforms. The configuration sets of patch size, encompassing small, medium, and large scales, have been established to cater to distinct feature extraction requirements at varying levels. The set of possible values is illustrated in Fig. 9 and includes {4, 8, 16}, {8, 16, 24}, and {16, 24, 32}. The configuration {8, 16, 24} demonstrated optimal performance across all datasets, with 86.80% on the MUUFL dataset and 95.29% on the Houston dataset. In contrast, both the excessively small patch size configuration {4, 8, 16} and the excessively large configuration {16, 24, 32} exhibited varying degrees of performance degradation. In light of this, CSS-Mamba adopts the patch size configuration of {8, 16, 24}.

2) *Learning Rate and Batch Size:*

In order to ascertain the optimal parameter combination of learning rate and batch size for the model, an experimental evaluation was conducted to determine the impact of the learning rate set $\{1 \times 10^{-5}, 5 \times 10^{-5}, 5 \times 10^{-4}, 5 \times 10^{-3}\}$ and the batch size set {16, 32, 64, 128} on OA, as illustrated in Fig. 10. As illustrated, the shaded bar regions indicate the parameter combinations that yield the highest OA, while the white bar regions denote those with the lowest OA values. For the MUUFL dataset, it has been demonstrated that larger batch sizes and smaller learning rates generally yield superior results. Conversely, moderate batch sizes paired with smaller learning rates prove more suitable for the Houston 2013 dataset. Moreover, the Trento dataset demonstrates that the batch size has a negligible effect on outcomes, while smaller learning rates generally yield superior overall accuracy. It is evident from the experiments conducted that the optimal learning rate and batch size for the CSS-Mamba framework are 5×10^{-5} and 64, respectively.

3) *Group size G:*

In the MS-SSSE module, the cluster size G, utilized for spectral subspace decomposition, dictates the granularity at which spectral features are decomposed into independent subspaces. This, in turn, exerts a direct influence on the strength of the structural sparsity constraint and the discriminative power of the features. As illustrated in Table VII, a comparative analysis has been conducted on four configurations ranging from $G = B/8$ to $G = B$ (no clustering) across three datasets. It was demonstrated that $G = B/4$ achieved the best performance across all three datasets. Following the reduction of G to $B/8$, an excessive number of spectral channels were observed within each subspace. This resulted in the spectral response patterns of different materials overlapping within the same subspace. This resulted in the weakness of the subspace decomposition's

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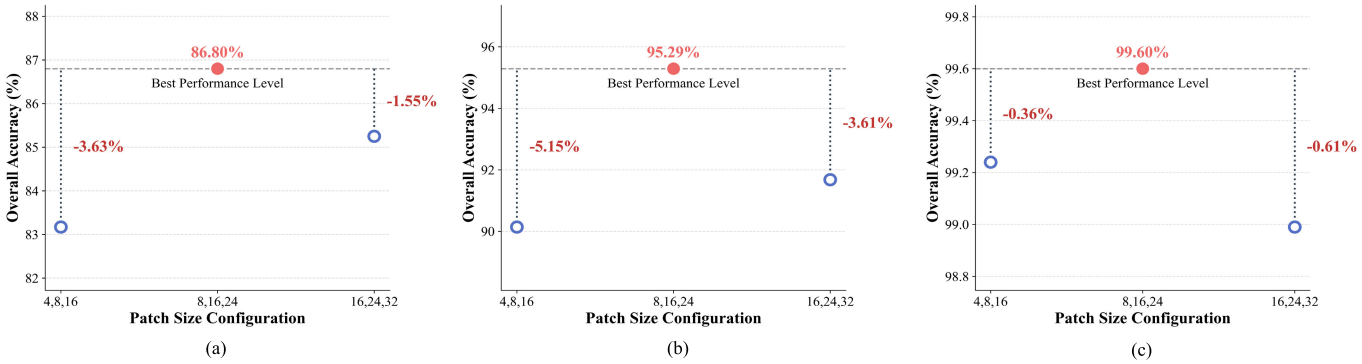


Fig. 9. Effect of input patch size. (a) MUUFL dataset. (b) Houston 2013 dataset. (c) Trento dataset.

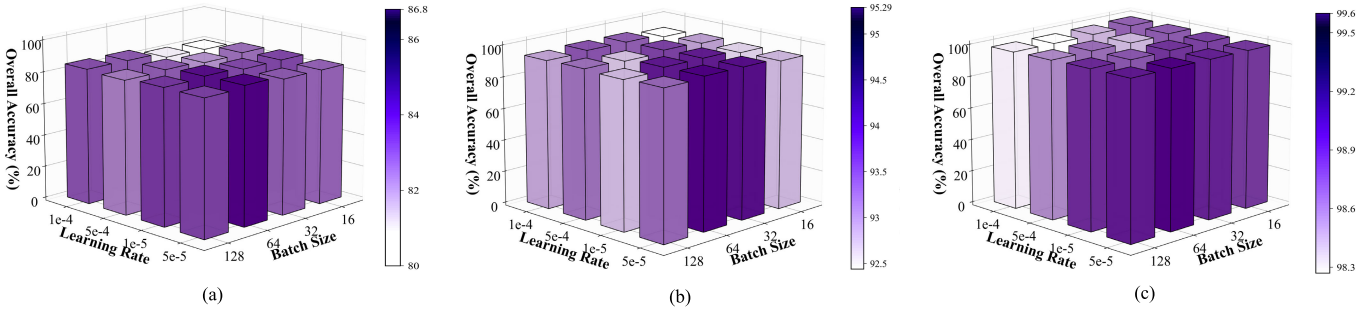


Fig. 10. Impact of varying learning rates and batch sizes on overall accuracy (OA). (a) MUUFL dataset. (b) Houston 2013 dataset. (c) Trento dataset.

TABLE VII
PARAMETER SENSITIVITY ANALYSIS OF GROUP SIZE G

Grouping Strategy	Houston OA(%)	MUUFL OA(%)	Trento OA(%)
$G = B/8$	94.71	85.43	99.21
$G = B/4$	**95.29**	**86.80**	**99.60**
$G = B/2$	94.88	86.12	99.38
$G = B$	94.15	85.67	99.07

ability to capture fine-grained spectral differences, which led to varying degrees of performance degradation across all three datasets. It is evident that as the value of G is increased to $B/2$ or even B, the structural sparsity constraint undergoes a gradual dissipation. It is evident that, at $G = B$, grouped convolution undergoes a transformation that leads to its degeneracy into a standard 1D convolution. This process results in the complete loss of the subspace structure, which consequently leads to a substantial decline in performance, a phenomenon that is particularly pronounced when analyzing the MUUFL dataset, which possesses a lower spectral dimension.

4) *Directional Strip Convolutional Kernels:*

To systematically evaluate the impact of asymmetric kernel dimensions in the MS-GSSE module, five configurations are compared: Config-A (1, 3)/(3, 5)/(5, 7), Config-B (3, 3)/(5, 5)/(7, 7), Config-C (3, 5)/(5, 9)/(7, 13), Config-D (5, 9)/(7, 13)/(9, 17), and Config-E (7, 13)/(9, 17)/(11, 21). The experimental results are displayed in Fig. 11. Config-A yields the lowest OA across all three datasets, with the most pronounced degradation on the Houston dataset, indicating insufficient receptive field coverage for dense linear

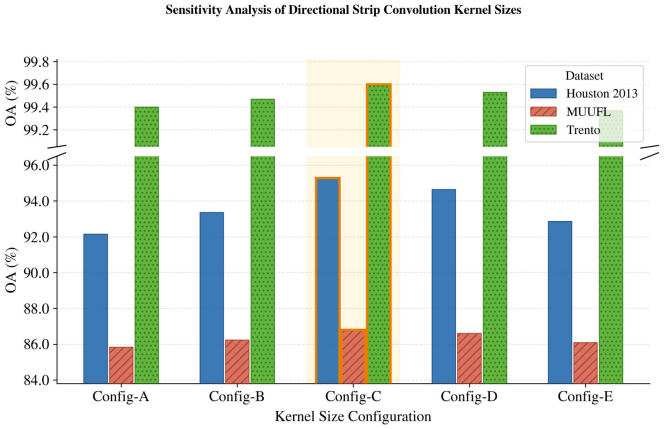


Fig. 11. Sensitivity analysis of directional strip convolution kernel size configurations on three benchmark datasets.

structures. Config-B, which removes directional asymmetry by setting $k_1 = k_2$, shows marginal improvement over Config-A but remains at a low-to-moderate level, confirming that asymmetric design is necessary for directional feature capture. Config-D achieves performance comparable to Config-C, suggesting negligible marginal gains from further expanding the receptive field. Config-E produces the weakest overall results among the five configurations, with the largest performance drop on the Houston dataset. Config-C achieves the best results across all three datasets, as its progressively expanding kernel sizes across scales enable an optimal balance between receptive field coverage and directional modeling capability.

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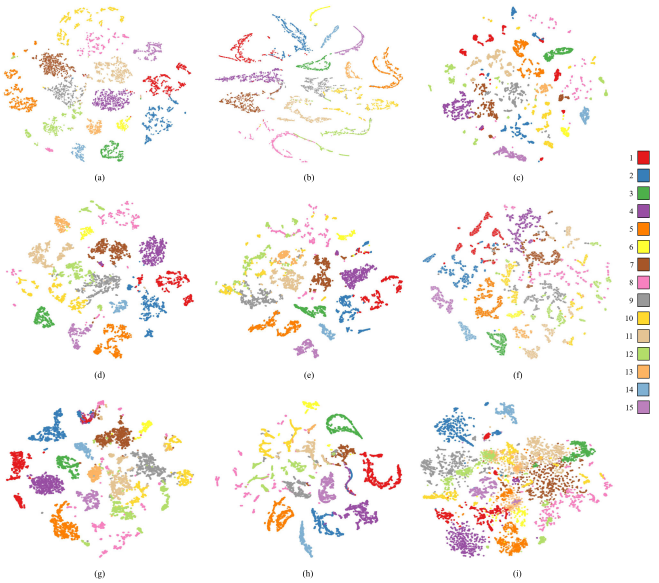


Fig. 12. t-SNE visualization of learned features for the Houston 2013 dataset. (a) CSS-Mamba, (b) DAHGMN, (c) MSFMamba, (d) MS2CANet, (e) MICF-Net, (f) HCT, (g) Cross-HL, (h) GAMF, (i) ExViT.

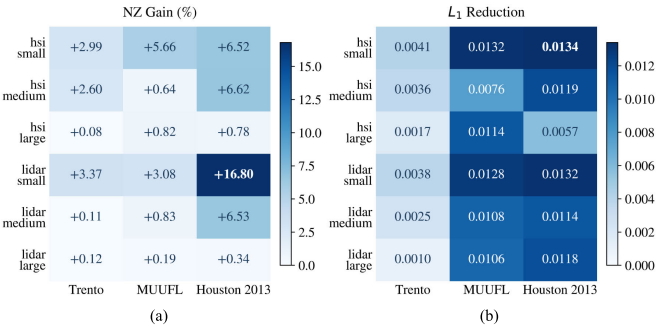


Fig. 13. NZ Gain (%) and L_1 Reduction heatmaps of soft-thresholding effects across all encoding modules on three datasets. (a) NZ Gain (%) heatmap. (b) L_1 Reduction heatmap. NZ Gain = NZ After − NZ Before reflects the improvement in sparsity, and L_1 Reduction = L_1 Before $-L_1$ After reflects the suppression of overall activation magnitude. Darker cells indicate stronger sparsification effects.

F. Feature Visualization Using t-Distributed Stochastic Neighbor Embedding (t-SNE)

The evaluation of the feature representation capability of the proposed CSS-Mamba framework is achieved by employing t-SNE for the purpose of visualizing feature distributions in a 2-D space. As illustrated in Fig. 12, the proposed model is presented alongside various state-of-the-art methods on the Houston 2013 dataset, employing t-SNE visualization. It is worth noting that all feature representations used for t-SNE visualization are extracted exclusively from the test set samples. Each data point is representative of a sample feature, with its color corresponding to the respective category label. The analysis demonstrates that the features extracted by CSS-Mamba manifest as pronounced intraclass clustering and interclass separation within the low-dimensional space. In general, there is a reduced number of isolated feature points, with features of the same category distributed more compactly and densely. Conversely, methods with lower classification

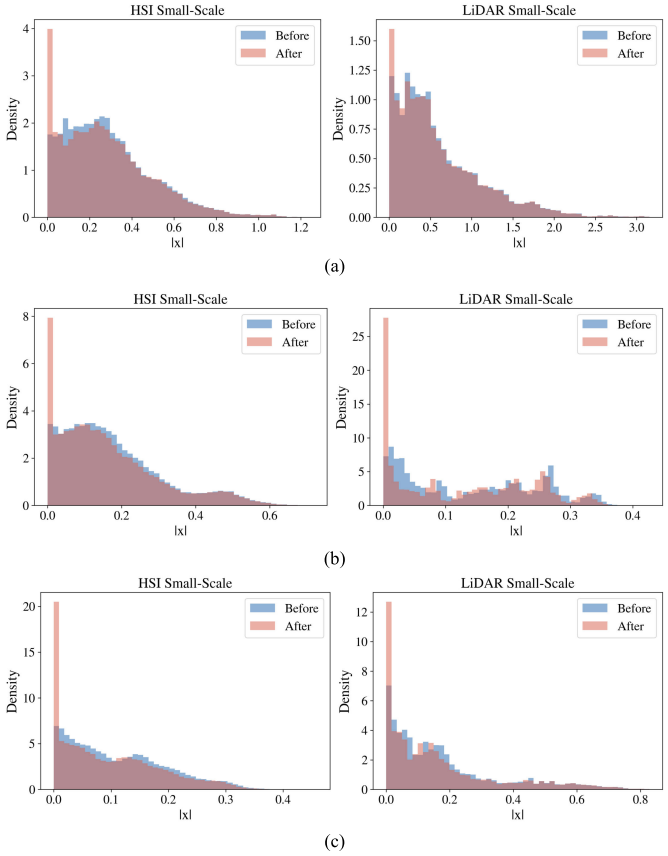


Fig. 14. L1 norm distribution histograms before and after soft-thresholding for the most representative module on each dataset (blue: before thresholding; red: after thresholding). (a) MUUFL dataset. (b) Houston 2013 dataset. (c) Trento dataset.

accuracy (e.g., Cross-HL and ExViT) demonstrate more dispersed feature distributions, where the boundaries between different category features appear blurred. This feature distribution pattern fully validates CSS-Mamba's advantage in effectively integrating multimodal data, constructing feature representations for objects that are highly discriminative and possess superior separability.

G. Sparsity of Soft Thresholding Mechanisms

To quantify the sparsity constraints imposed by the learnable soft-thresholding operator, we extract feature maps immediately before and after each thresholding layer from a trained model during inference and compare their statistics across all six encoding modules on three datasets. Near-zero elements are defined as those with absolute value below 0.01. A sparse feature map concentrates its energy in a small subset of elements, while the majority approaches zero; this structural property manifests as a simultaneous increase in the near-zero ratio and a decrease in the mean L_1 norm. As shown in Fig. 13, both effects are consistently observed across all modules and datasets. The NZ Gain heatmap reveals that sparsity improvements are most pronounced in small-scale modules and on the Houston 2013 dataset, reaching up to 16.80 percentage points, while large-scale modules exhibit comparatively modest gains. The L_1 Reduction heatmap shows

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TABLE VIII

COMPARISON OF COMPUTATIONAL COMPLEXITY AND TRAINING EFFICIENCY AMONG DIFFERENT METHODS ON THE HOUSTON 2013 DATASET. PARAMS: NUMBER OF LEARNABLE PARAMETERS (M); FLOPS: FLOATING-POINT OPERATIONS PER SAMPLE (M); AND TRAIN TIME: AVERAGE TRAINING TIME PER EPOCH (S/EPOCH)

Metrics	ExViT	HCT	GAMF	Cross-HL	MICF-Net	MS2CANet	MSFMamba	DAHGMN	CSS-Mamba
Params (M)	0.24	0.45	6.96	0.28	0.18	0.90	1.08	0.35	0.80
FLOPs (M)	186.85	16.08	2946.49	44.52	20.09	15.98	196.66	93.12	145.60
Train Time (s/epoch)	0.36	0.19	2.98	0.15	0.56	0.19	0.17	0.15	0.33

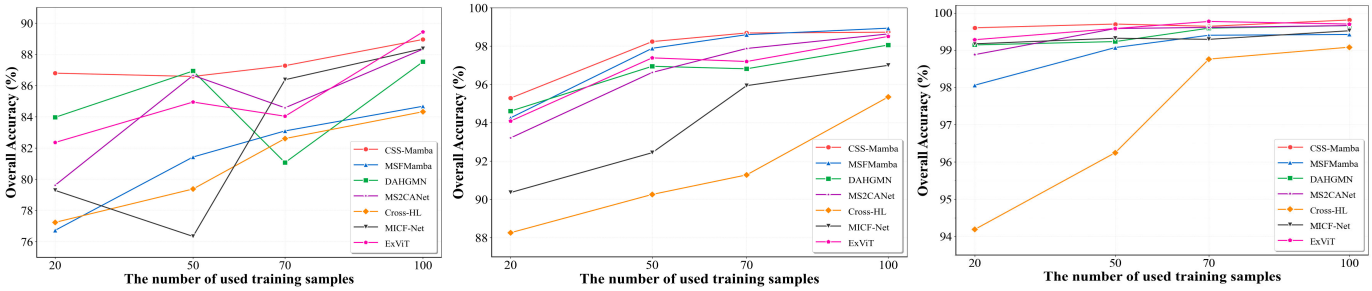


Fig. 15. Impact of different methods using varying training sample sizes on the overall accuracy (OA). (a) MUUFL dataset. (b) Houston 2013 dataset. (c) Trento dataset.

a consistent pattern across modalities, with the highest reductions concentrated in the Houston 2013 column. The L_1 norm distribution histograms in Fig. 14, which display the modules with the largest NZ Gain on each dataset, further illustrate how this sparsification occurs: the postthresholding distributions (red) show elevated near-zero density peaks, while high-activation regions remain largely unaffected, confirming that weak responses are selectively suppressed rather than features being uniformly compressed.

H. Analysis of Computational Complexity

To conduct a comprehensive analysis of computational complexity, the model parameters, floating-point operations per second (FLOPs), and training time per epoch for different methods on the Houston 2013 dataset were evaluated, the results of which are shown in Table VIII. In summary, CSS-Mamba attains competitive classification accuracy while exhibiting minimal parameters and training overhead. In terms of parameter count, CSS-Mamba contains only 0.80 million trainable parameters, which is significantly lower than GAMF (6.96 million) and MSFMamba (1.08 million), and is comparable to other lightweight models. With regard to computational cost, the FLOPs of CSS-Mamba are 145.60M, which is lower than those of MSFMamba (196.66M) and particularly advantageous in comparison to those of GAMF (2946.49M). With regard to the time allotted to training, CSS-Mamba requires 0.33 s for each epoch, maintaining a reasonable training duration among the evaluated methods while demonstrating a time advantage over GAMF (2.98 s/epoch). A combined analysis of the three metrics reveals that CSS-Mamba attains the maximum classification accuracy among all the methods that have been evaluated, given their moderate parameter counts, computational complexity, and training overhead. This finding serves to substantiate the efficacy of the proposed method, which exhibits an optimal equilibrium between efficiency and performance.

I. Comparison Under Different Training Sample Sizes

The classification accuracy of a model is found to be significantly influenced by the scale of the training dataset. In order to analyze the performance of the proposed CSS-Mamba under varying sample sizes, experiments were designed with incrementally increasing training samples. The number of samples per class was set to 20, 50, 70, and 100, respectively. In order to ensure experimental fairness, all methods employed an identical sample allocation strategy. As illustrated in Fig. 15, the OA of various classification methods is demonstrated across a range of sample sizes. The OA curves for each method demonstrate a consistent upward trend with increasing training data volume. It is noteworthy that the proposed CSS-Mamba not only maintains competitiveness in scenarios with ample samples but also demonstrates particularly outstanding performance in limited-sample settings, fully showcasing its adaptive advantage for small-sample data.

V. CONCLUSION

This article presented a CSS-Mamba for hyperspectral–LiDAR joint classification, aiming to explicitly model modality-specific sparsity and alleviate redundancy in multiscale feature fusion. Unlike conventional multimodal fusion approaches that adopt homogeneous feature extraction and implicit aggregation, the proposed framework integrates modality-aware sparse encoding with competitive multiscale feature selection within a state-space modeling architecture, enabling effective exploitation of the complementary spectral and geometric characteristics of hyperspectral and LiDAR data. By designing dedicated multiscale sparse encoding modules for hyperspectral imagery and LiDAR data, CSS-Mamba

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explicitly captures spectral sparsity and geometric sparsity in a modality-differentiated manner. Furthermore, the proposed CSS mechanism establishes explicit competition among multiscale features, adaptively enhancing discriminative responses while suppressing redundant ones. This design effectively improves feature representation quality in complex scenes without introducing excessive computational burden.

Extensive experimental results on three widely used benchmark datasets—Trento, Houston 2013, and MUUFL—demonstrate that CSS-Mamba consistently outperforms representative CNN-, Transformer-, and Mamba-based multimodal fusion methods in terms of overall accuracy, average accuracy, and Kappa coefficient. The results confirm the robustness and generalization capability of the proposed framework across different data characteristics and scene complexities. In future work, we plan to extend the proposed framework to more challenging multimodal remote sensing scenarios, such as large-scale urban mapping and multitemporal data fusion, and to further explore lightweight model designs for efficient deployment in resource-constrained environments.

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